

Name		Class	9th	Marks	60
Roll No#		Subject	Physics	Time	2 Hours
Syllabus	Units 1, 2, 3, 4	Date	/ /	Obt. Marks	
COMPREHENSIVE PHYSICS TEST					

Q1: Choose the correct option.

(12)

- (i) One femto meter is equal to:
- A) 10^9 m B) 10^{-15} m C) 10^{-9} m D) 10^{15} m
- (ii) Which one is a non-physical quantity?
- A) Distance B) Density C) Colour D) Temperature
- (iii) Identify the base quantity in the following:
- A) Volume B) Density C) Current D) Force
- (iv) The numerical ratio of displacement to distance is:
- A) Always less than 1 C) Always greater than 1
B) Always equal to 1 D) Equal to or less than 1
- (v) A ball is dropped from the top of a tower, the distance covered by it in the first second is:
- A) 5 m B) 10 m C) 50 m D) 100 m
- (vi) The area under the speed-time graph is numerically equal to:
- A) Velocity B) Uniform velocity C) Acceleration D) Distance covered
- (vii) The mass of the body is a measure of its:
- A) Inertia B) Weight C) Rest D) Acceleration
- (viii) The product $F \times \Delta t$ is called:
- A) Impulse B) Momentum C) Torque D) Acceleration
- (ix) A lubricant is usually introduced between two surfaces to decrease friction. The lubricant:
- A) Decreases temperature C) Prevents direct contact
B) Acts as ball bearing D) Provides rolling friction
- (x) A force F is making an angle of 60° with x-axis, its y-component is equal to:
- A) F B) $F \sin(60^\circ)$ C) $F \cos(60^\circ)$ D) $F \tan(60^\circ)$
- (xi) The center of mass of a body:

- A) Lies always inside
- B) Lies always outside

- C) Lies always on the surface
- D) May lie within, outside or on surface

(xii) Moment of force is called:

- A) Momentum
- B) Couple
- C) Couple arm
- D) Torque

SUBJECTIVE PART

Q2: Write short answers to any FIVE (5) questions. (10)

- (i) Why a standard unit is needed to measure a quantity correctly?
- (ii) Write the name of 3 base quantities and 3 derived quantities.
- (iii) Define scalar and vector quantities. Give examples.
- (iv) What are distance-time graph and speed-time graph?
- (v) Define impulse and force.
- (vi) State Newton's 2nd law of motion.
- (vii) Define like and unlike parallel forces.
- (viii) What are rectangular components of a vector?

Q3: Write short answers to any FIVE (5) questions. (10)

- (i) What is measurement? Name its two parts.
- (ii) State the number of significant digits in 0.0045 m and 2.047 m.
- (iii) Is it possible that the velocity of an object is zero at an instant of time but its acceleration is not zero? Give an example.
- (iv) A stone dropped from a height of 45 m. How long will it take to reach the ground?
- (v) Why has not Newton's first law been proved on the Earth?
- (vi) Action and reaction are always equal and opposite then how does a body move?
- (vii) Give an example of a body which is moving yet in equilibrium.
- (viii) Why do we put the handle of a door at its extreme edge?

Q4: Write short answers to any FIVE (5) questions. (10)

- (i) Why is it better to place the object close to the meter scale?
- (ii) Solve: $5.123 \times 10^4 \text{ m} + 3.28 \times 10^5 \text{ m}$.
- (iii) Falling objects near the earth have the same constant acceleration. Does this imply that a heavier object will fall faster than a lighter object?
- (iv) An eagle dives to the ground along a 300 m path with an average speed of 60 m/s. How long does it take?
- (v) When a cricket ball hits high, a fielder tries to catch it. While holding the ball he/she draws hands backward. Why?
- (vi) When someone jumps from a small boat onto the river bank, why does the jumper often fall into the water?
- (vii) What is center of mass? Explain its effect on rotation.

(viii) Can a body be in equilibrium if it is revolving clockwise under the action of a single force?

Note: Write detailed answers to any TWO (2) questions.

(9+9=18)

Q5: (a) What is meant by base and derived quantities? Give symbols of SI base units. (4)

(b) Prove that the area under a speed-time graph is equal to the distance covered by an object.

(5)

Q6: (a) State and explain the principle of conservation of momentum. (4)

(b) A 5 N ball hanging from a rope is pulled to the right by a horizontal force F . The rope makes an angle of 60° with the ceiling. Determine the magnitude of force F and tension T in the string. (5)

Q7: (a) State and explain the two conditions of equilibrium. (4)

(b) A 5 g bullet is fired by a gun. The bullet moves with a velocity of 300 m/s. If the mass of the gun is 10 kg, find the recoil speed of the gun. (5)